

google colab - Searchexp 1.ipynb - ColabUntitled7.ipynb - Colab

https://colab.research.google.com/drive/1y4XEKjxn05athl479vAIG7J5ctRM#scrollTo=1f5jSnXDYBv

CommandsCodeTextRun allRAMDisk

[6]

```
# Training data
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]
# Initialize S and G
S = data[0][:-1]
G = [['?' for _ in range(len(S))]]
# Algorithm
for row in data:
    x = row[:-1]
    y = row[-1]
    if y == 'Yes': # Positive
        for i in range(len(S)):
            if S[i] != x[i]:
                S[i] = '?'
    else: # Negative
        new_G = []
        for g in G:
            for i in range(len(S)):
                if g[i] == '?' and S[i] != x[i]:
                    h = g.copy()
                    h[i] = S[i]
                    new_G.append(h)
        G = new_G
# Output
print("Final S:", S)
print("Final G:", G)
```

Final S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
Final G: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', 'Same']]

VariablesTerminal

7:37 PMPython 3

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